The empirical recurrence for OEIS A263971: recovering a conjecture that is not written down

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Abstract

OEIS A263971 records "Empirical recurrence of order 53", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 1024$ vertices and satisfies some monic recurrence of order at most S; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 53, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 53 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A263971 is "Number of (n+1)X(4+1) arrays of permutations of 0..n*5+4 with each element having index change (+-,+-) 0,2 1,1 or 1,0.". It has offset 1 and begins

$$465, 10720, 2604481, 226810352, 31702909989, 3594130469760, 448695451459909, 53640316831369632, \ldots$$

The entry states:

> Empirical recurrence of order 53 (see link above)

As of the "Last modified" line on the live entry (Nov 10 2015 10:29:14, revision 6) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1024$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 1024$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2048$ exact terms of the model returns order 53 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{53} c_i \, a(n-i),$$

c_1	100	c_{15}	−435250675888896056688	c_{29}	78986562369395514289567662080
c_2	6087	c_{16}	161818387177195861296	c_{30}	−6706712031173755395816226816
c_3	−390976	c_{17}	475912773522818875440	c_{31}	−534621920393806037643865751552
c_4	−8403494	c_{18}	−1486483114546636158032	c_{32}	989489981213535893481084157952
c_5	474196768	c_{19}	17935195481479563200224	c_{33}	−668221624829111426184466923520
c_6	5002152626	c_{20}	−145444951281278549025344	c_{34}	−1660479584643431598263078223872
c_7	−260715013792	c_{21}	−249150635744669666536320	c_{35}	7190243724787351229119584010240
c_8	−1403473040977	c_{22}	3076760846040317647500032	c_{36}	−5096317403500266722222499954688
c_9	74204170986356	c_{23}	−4785223727969525937227264	c_{37}	−775947942823691568513814626304
c_{10}	138338337533047	c_{24}	15408224934677544376771584	c_{38}	−9684858390747275318422101032960
c_{11}	−11747087671883792	c_{25}	−32145094419417547582588928	c_{39}	−1384504055891744739328585105408
c_{12}	8825649132981656	c_{26}	129398530007428453999210496	c_{40}	22414394229124014782591520473088
c_{13}	1022710783433343640	c_{27}	−357210748892431242891976704	c_{41}	6152966768214105487220402028544
c_{14}	−2637146528978763792	c_{28}	−15866489056824725565236297728	c_{42}	1009789285296792988461912031232

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 53, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $53+4$ terms removed returns order 53 as well, so $\deg q_{\min} = 53 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $53+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 53 that this sequence satisfies — and that family turns out to have exactly one member.

References

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